

CS07340 DESIGN AND ANALYSIS OF ALGORITHMS

WEEK 1 QUIZ (8 POINTS)

Name: _____

1. (4 points) In this problem you will count the exact number of steps for the code segment `Sum(int[] A)` below. (Do not count `main`) You should count each assignment (`=`, `+=`) and each comparison (but **not** the return statement) as one step. `A.length` is the variable we called n in class. Write your answer as a function of n (for example n steps or $\frac{1}{2}n^2$ steps).

```
public class Main {
    public static int Sum(int[] A) {
        int s = 0;
        for(int i = 0; i < A.length; i++) {
            s += A[i];
        }
        return s;
    }
    public static void main(String[] args) {
        int[] A = new int[]{1,2,3,4,5,6,7};
        System.out.println(Sum(A));
    }
}
```

|||||
1 +
7 multiple of $n = A.length$

@
5 steps

2. (4 points) The following are True/False questions about algorithm determinism. Clearly mark either $\textcircled{T}$ (for True) or $\textcircled{F}$ (for False).

- $\textcircled{T}$ ☒ $\textcircled{F}$ The problem of computing the number π exactly can be solved with an algorithm.
 $\textcircled{T}$ ☒ $\textcircled{F}$ The problem of computing the number π approximately can be solved with an algorithm.
 $\textcircled{T}$ ☒ $\textcircled{F}$ The problem of converting a number from decimal representation to binary representation can be solved with an algorithm.
 $\textcircled{T}$ ☒ $\textcircled{F}$ An algorithm may have infinitely many steps.